A Project Presentation

on

Speech Emotion Recognition for Early Detection of Depression and Mental Health Issues

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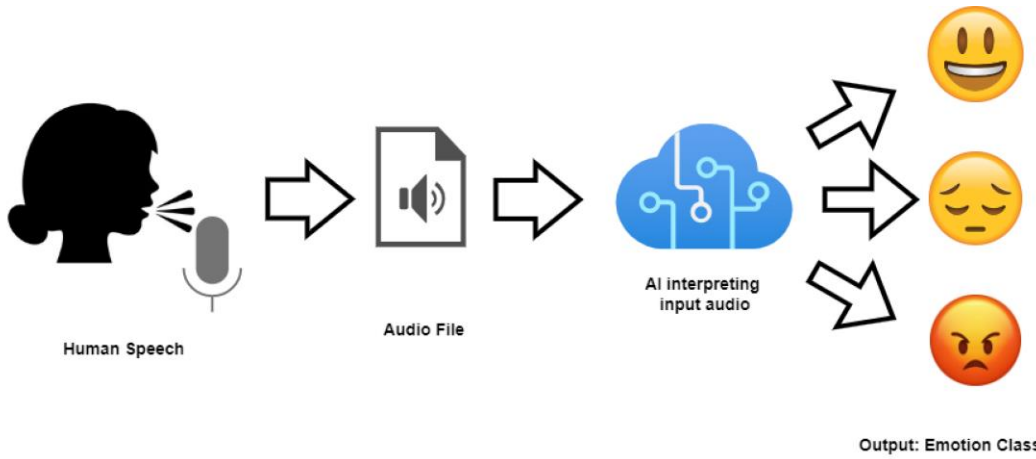
Introduction

- **Depression often remains undiagnosed** due to lack of visible or early signs.
- **People hesitate to express** their emotions or seek professional help.
- **Speech is a natural**, non-invasive medium to understand emotional state.
- **Traditional diagnosis** methods like interviews or surveys are slow and subjective.



Abstract

- Combines **speech emotion recognition** to detect early signs of depression and mental health issues.
- Classifies core emotions such as **sadness, anger, happiness, fear, and neutrality**.
- Provides a **real-time solution** for mental health monitoring.
- Aims to support **early detection and timely intervention** in depression and other mental disorders.





Objective

1. To preprocess speech datasets and extract depression-specific vocal features.
2. To develop a SER model for early detection of depression.
3. To compare ML and DL techniques.
4. To design a real-time emotion monitoring system.

Literature Survey

Sr. No	Title and Author Name	Key Idea	Advantages	Disdvantages
1	"Real-time Speech Emotion Recognition Using Deep Learning and Data Augmentation“ **Author:**Chawki Barhoumi, Yassine BenAyed	• Uses CNN & LSTM with data augmentation to detect emotions in real time from speech	• Real-time detection, better accuracy with augmentation.	• Needs high computing power, struggles in noisy environments.
2	Automatic Speech Emotion Recognition Using Machine Learning: Digital Transformation of Mental Health	• Uses ML models with MFCC & Chroma features to detect emotions for mental health monitoring.	• Recognizes emotions from speech, supports digital mental health checks.	• Limited dataset diversity, performance drops with noise.

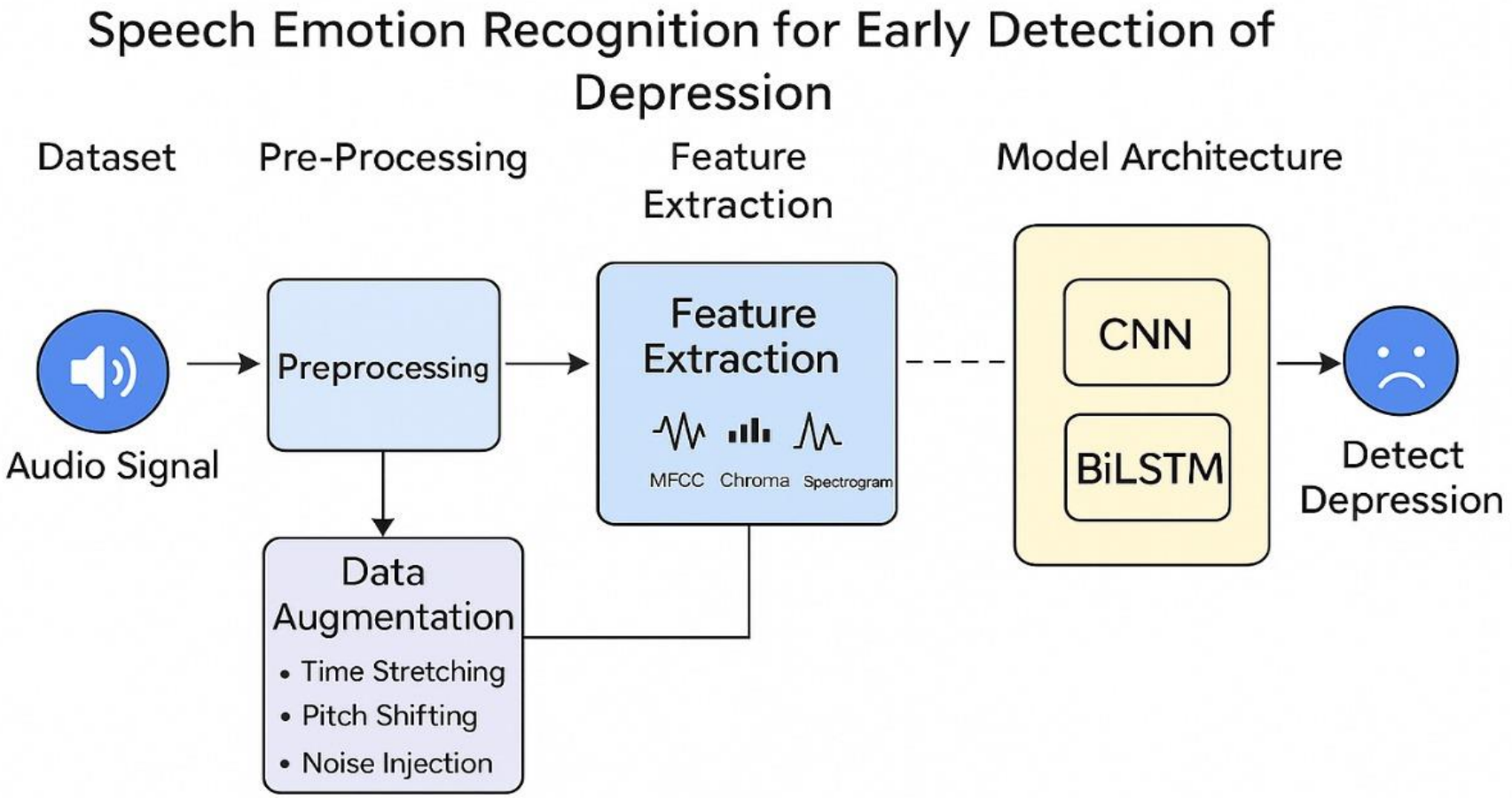
Literature Survey

Sr.No	Title and Author Name	Key Idea	Advantages	Disadvantage
3	"Multi-Modal Fused-Attention Network for Depression Level Recognition" Authors: Yihan Zhou and team (2023)	• Combines audio & video with attention to detect depression, focuses on emotional parts	• Accurate, catches subtle cues, good for hidden signs.	• Needs quality audio-video, not real-time, complex.
4	Multi-Level Attention Network Using Text, Audio, and Video" Authors: Anupama Ray and team (2019)	• Uses audio, video & text, finds key clues, detects mismatches in tone & words.	• 17.5% better accuracy, understands tone + words+ expressions.	• Needs all 3 data types, slow training, high resources.

Dataset

Parameter	Details
Dataset Name	RAVDESS
Total Actors	**24 (12 Male + 12 Female)**
Total Audio Files	**1,440 Speech Audio Files**
Audio Format	WAV
Sampling Rate	48 kHz (Resampled to 16 kHz)
Recording Quality	High-quality Studio Recorded
Emotion Classes	8
Total Statements	2
Emotional Intensity	Normal & Strong

System Architecture



Methodology

Step 01
Speech Input
Record live speech or upload audio.

Step 02
Preprocessing
Noise Reduction, Silence Removal, Normalization.

Step 03
Feature Extraction
MFCC Extraction, Chroma Feature Extraction, Mel Spectrogram Extraction

Step 04
CNN + LSTM Hybrid Model
Learn spatial emotional features. Capture temporal dependencies.

Step 05
Emotion Classification
Predict emotion using Softmax.

Step 06
Depression Risk
Estimate depression risk.

Step 08
Result
Display emotion, graphs and score.

Algorithms

1. MFCC (Mel-Frequency Cepstral Coefficients)

- Extracts important speech features.
- Converts speech into numerical vectors.
- Captures emotional characteristics.

2. CNN (Convolutional Neural Network)

- Learns spatial features from MFCC.
- Automatically extracts deep features.
- Improves classification accuracy.

3. BiLSTM (Bidirectional LSTM)

- Captures temporal dependencies.
- Processes forward & backward context.
- Enhances emotion recognition.

4. Softmax Classifier

- Calculates emotion probabilities.
- Predicts final emotion.
- Supports depression risk assessment.

Implementation

Technologies Used:

- Python
- TensorFlow / Keras
- Librosa
- Scikit-learn
- Matplotlib
- Seaborn

1. • Dataset Loading from RAVDESS
2. • Audio Augmentation :
   Noise Injection, Time Stretching and Pitch Shifting
3. • Feature Extraction
   MFCC, Chroma Feature, Mel spectrogram
4. • CNN-BiLSTM Model Training
5. • Emotion Classification into 8 Classes
6. • Depression Risk Prediction Based on Negative Emotions

Testing

Functional Testing

- Audio file upload
- Audio recording
- Emotion prediction
- Depression risk analysis

Integration Testing

- Audio preprocessing with feature extraction
- Feature extraction with CNN-BiLSTM model

System Testing

- Complete end-to-end workflow validation

Performance Testing

- Model accuracy evaluation
- Response Time analysis

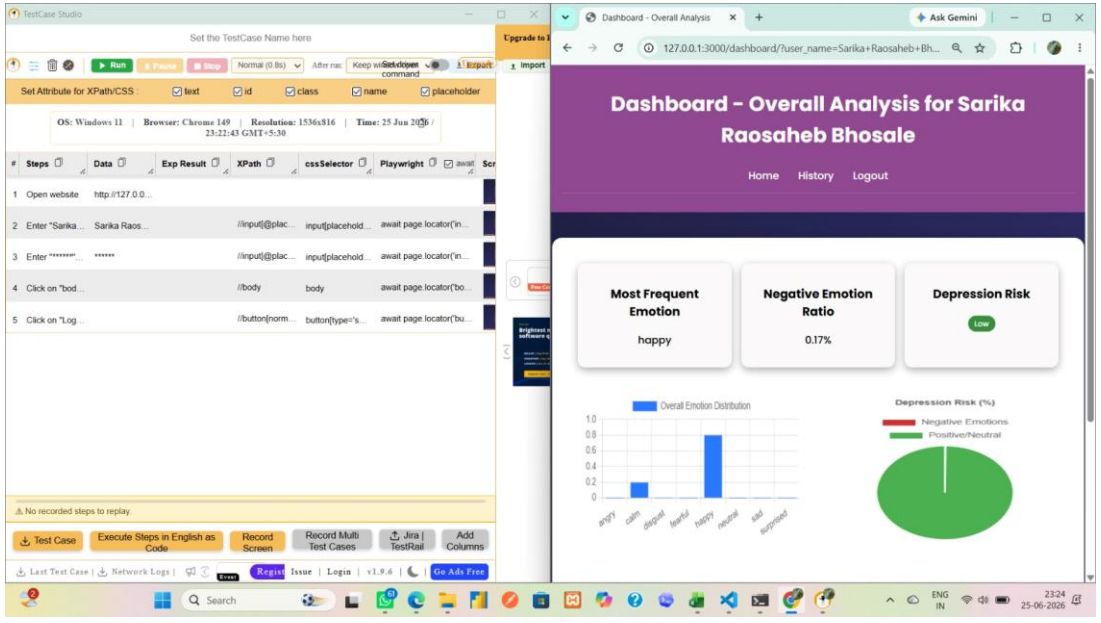
Testing Outcome

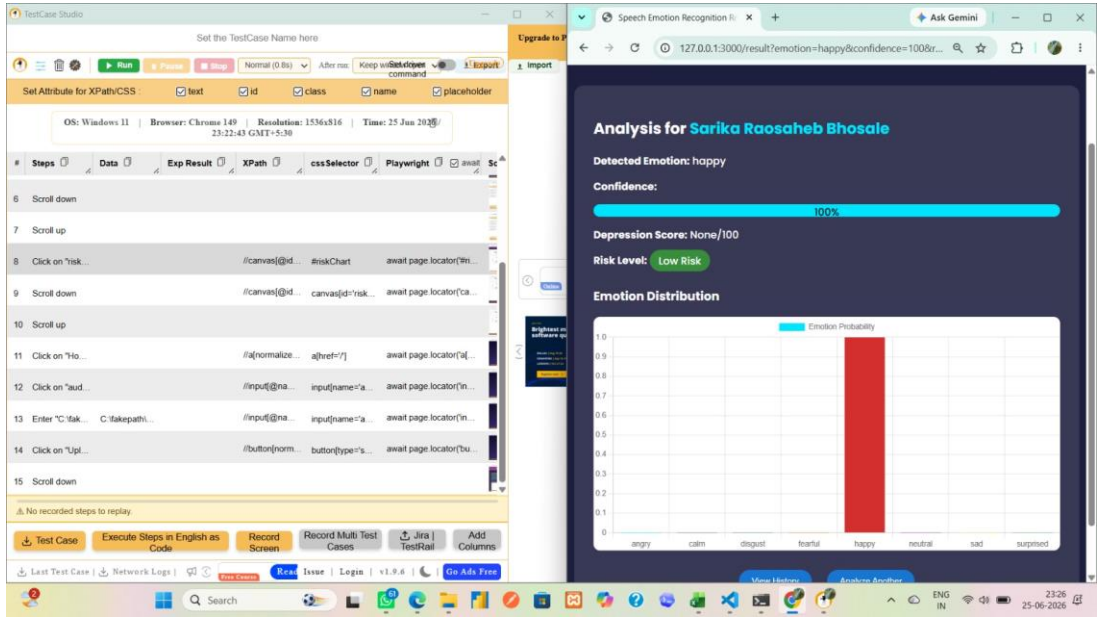
✓ All modules executed successfully
✓ Accurate emotion classification achieved
✓ Depression risk analysis generated correctly
✓ Stable system performance with valid audio inputs
✓ No critical defects found during testing

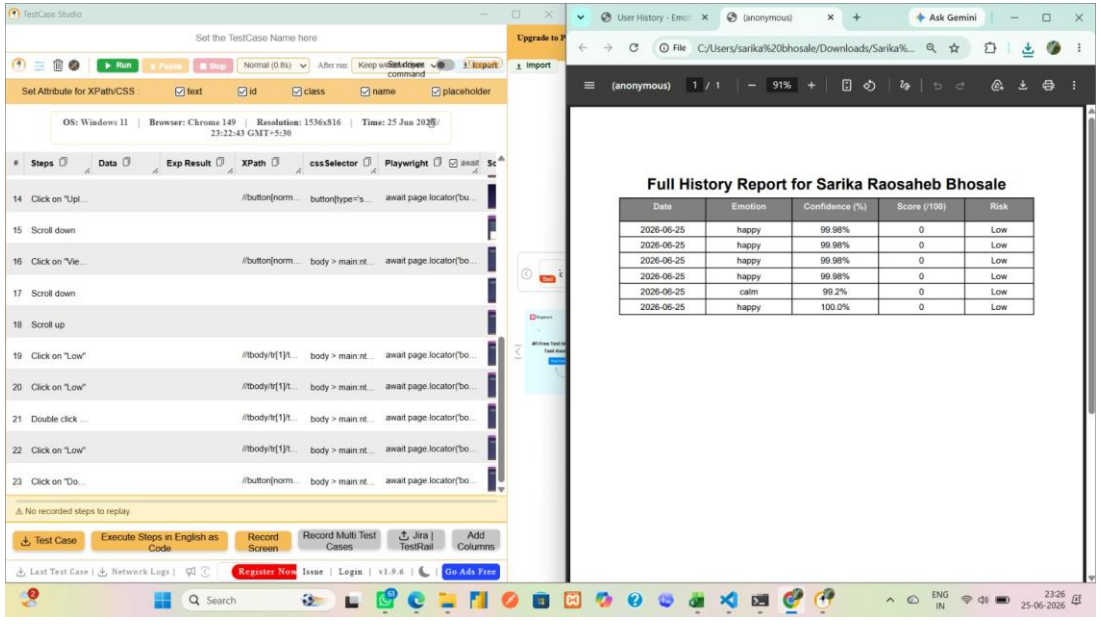
SELENIUM TEST CASES

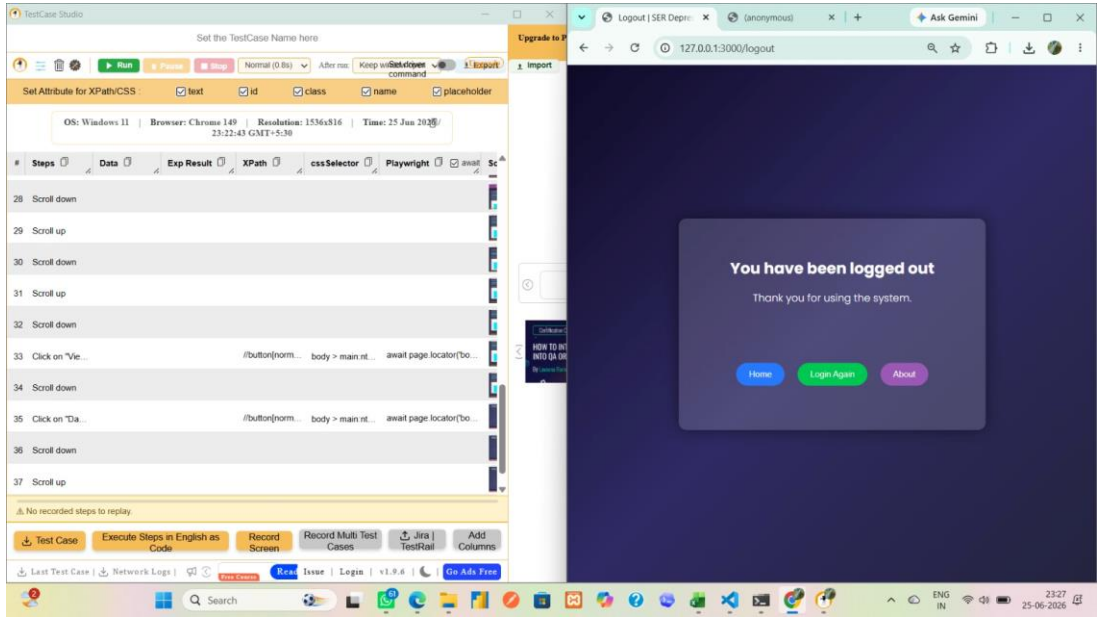
TC ID	Selenium Test Scenario	Expected Result	Status
AT01	Launch the Speech Emotion Recognition application	Application homepage loads successfully	Pass
AT02	Verify audio file upload button	Upload button is visible and clickable	Pass
AT03	Upload a valid WAV audio file	Audio file uploads successfully	Pass
AT04	Click the Predict Emotion button	Emotion prediction process starts successfully	Pass
AT05	Verify predicted emotion display	Predicted emotion is displayed correctly	Pass
AT06	Verify confidence score display	Prediction confidence score is displayed	Pass
AT07	Verify depression risk analysis output	Depression risk level is displayed correctly	Pass
AT08	Verify handling of invalid file format	Appropriate error message is displayed	Pass
AT09	Verify application reset/clear functionality	Uploaded data and results are cleared successfully	Pass
AT10	Perform complete end-to-end	Entire workflow executes	Pass

Testing Result

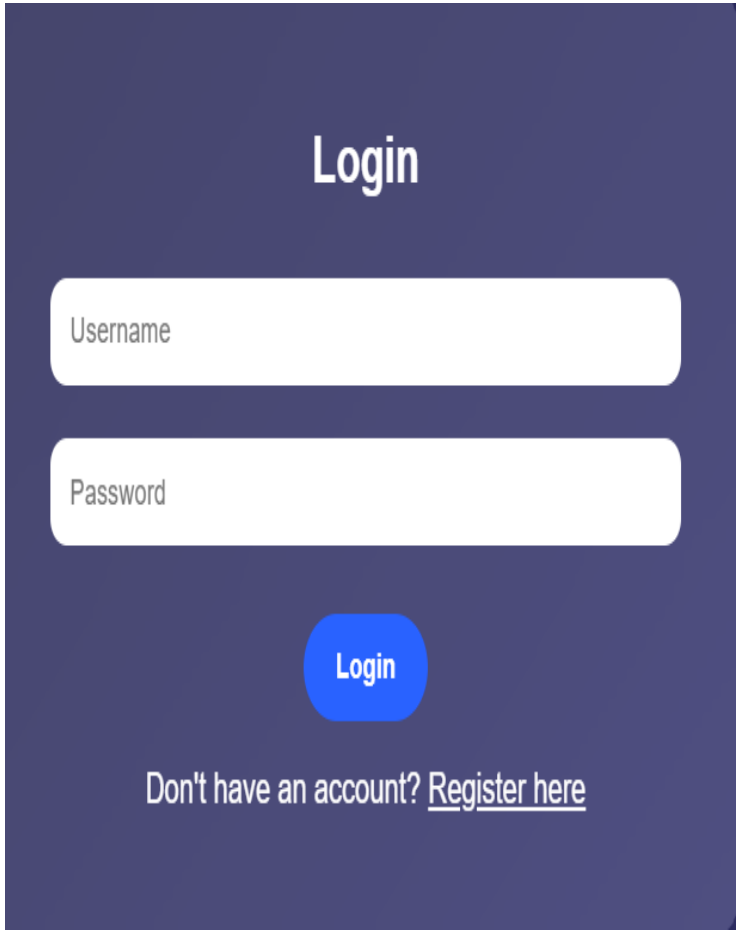




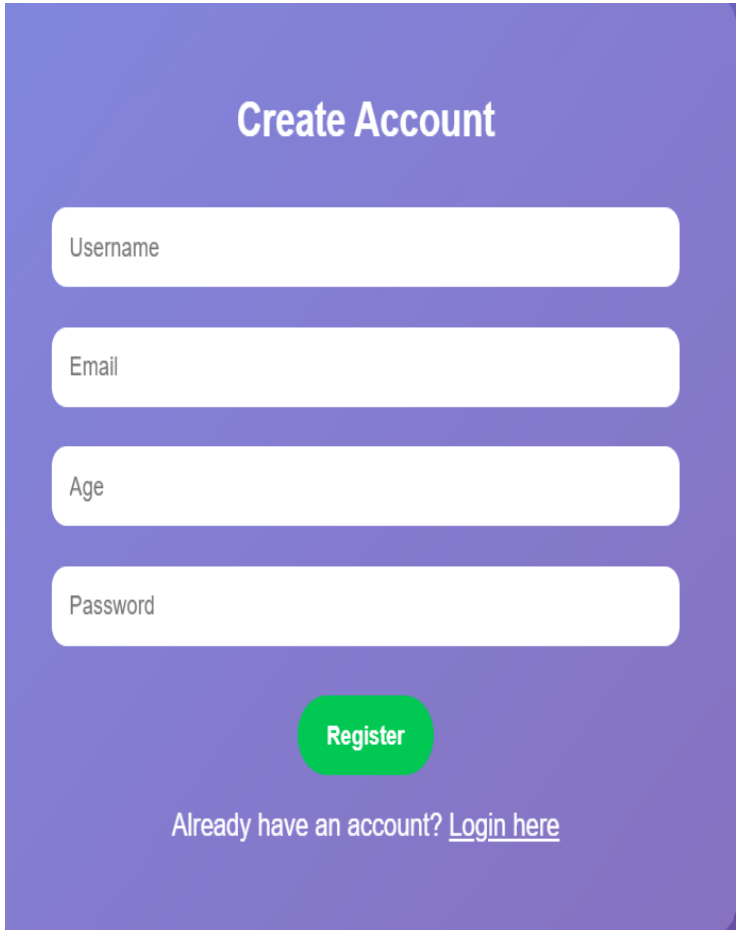




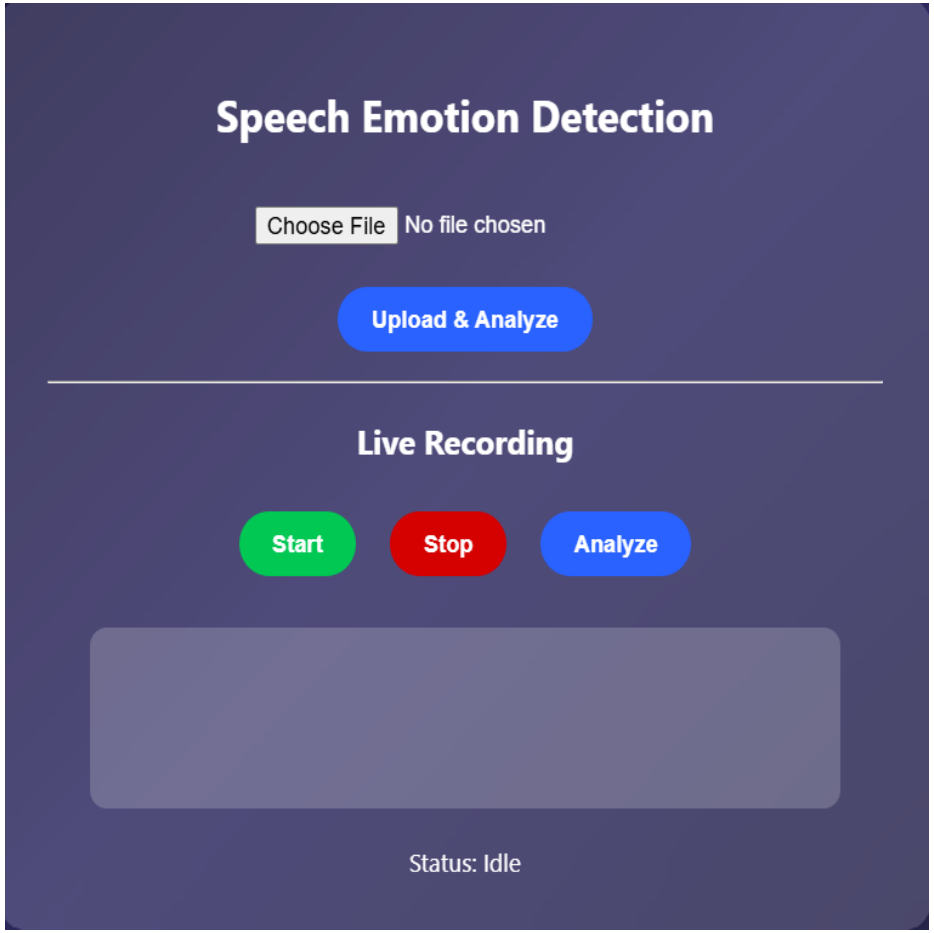
Result



Login Page



Registration

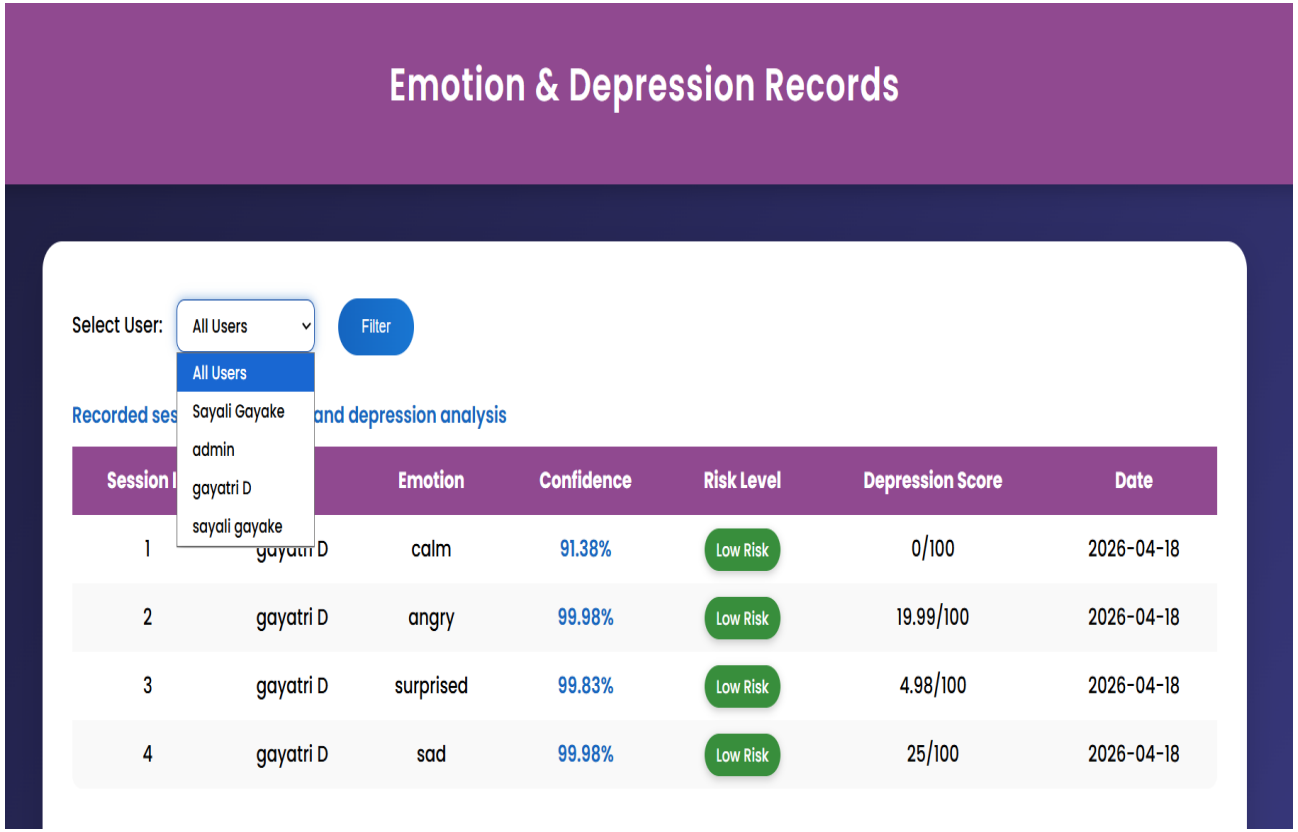


Audio Upload and Analysis Interface

Result



Dashboard



Emotion and Depression Records

Result

User History - admin

Session ID	Date	Emotion	Confidence	Depression Score	Risk Level	Source
1	2026-04-17	fearful	99.97%	14.99/100	Low	
2	2026-04-17	angry	100.0%	20/100	Low	
3	2026-04-17	surprised	99.97%	5.01/100	Low	
4	2026-04-17	angry	100.0%	20/100	Low	
5	2026-04-17	disgust	100.0%	25/100	Low	
6	2026-04-17	disgust	100.0%	25/100	Low	
7	2026-04-18	happy	99.96%	0/100	Low	
8	2026-04-18	happy	99.96%	0/100	Low	
9	2026-04-18	happy	99.96%	0/100	Low	

History

Full History Report for admin

Date	Emotion	Confidence (%)	Score (/100)	Risk
2026-06-25	calm	99.02%	0	Low
2026-06-25	calm	99.02%	0	Low
2026-06-25	calm	99.02%	0	Low
2026-06-25	calm	99.02%	0	Low
2026-06-25	calm	99.02%	0	Low
2026-06-25	calm	99.89%	0	Low
2026-06-25	calm	99.63%	0	Low
2026-06-25	calm	99.63%	0	Low
2026-06-25	calm	99.63%	0	Low
2026-06-25	calm	99.63%	0	Low
2026-06-25	calm	99.63%	0	Low
2026-06-25	calm	99.63%	0	Low

Downloaded Report

Conclusion

Conclusion

- Successfully processed speech audio from the RAVDESS dataset
- Extracted MFCC, Chroma, Delta, Delta-Delta, and Mel Spectrogram features
- Implemented a hybrid CNN–BiLSTM architecture for emotion classification
- Accurately recognized 8 emotional states
- Performed depression risk analysis based on detected emotions
- Achieved reliable performance with good accuracy and generalization

References

- **Research Papers**

1.Livingstone, S. R., & Russo, F. A. (2018). *The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A Dynamic, Multimodal Set of Facial and Vocal Expressions in North American English.* PLoS ONE.

2.Trigeorgis, G., et al. (2016). ***Adieu Features?*** *End-to-End Speech Emotion Recognition Using a Deep Convolutional Recurrent Network.* IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).

3.Neumann, M., & Vu, N. T. (2017). *Attentive Convolutional Neural Network Based Speech Emotion Recognition.* INTERSPEECH.

4.Satt, A., Rozenberg, S., & Hoory, R. (2017). *Efficient Emotion Recognition from Speech Using Deep Learning on Spectrograms.* INTERSPEECH.

- **Dataset Reference**

RAVDESS Dataset : Livingstone, S. R., & Russo, F. A. (2018).
Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS).

Thank You